Terrence Neumann

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About Me

I'm a PhD student researching ethical AI. I have extensive experience collaborating with academics, policy makers, and practitioners to use data for the greater good. My research incorporates machine learning, ethics, human-computer interaction, and social science.

Education

University of Texas at Austin **Austin, TX**
PhD – Information Systems, Ethical AI Emphasis *Expected 2026*

Northwestern University **Evanston, IL**
MS in Analytics *December 2016*

Indiana University, Bloomington **Bloomington, IN**
BA in Economics and Mathematics (Departmental Honors) *May 2015*

Professional Experience

University of Texas at Austin **Austin, TX**
Research Assistant – Good Systems Initiative *July 2021 - Present*

- Fine-tuned language model (BERT) and studied one-shot learning applications of AI for fact-checking.
- Developed philosophical notions of fairness in misinformation detection. Published at ACM FAccT '22.
- Designed and implemented a simulation of a blogging platform called `TACIT`, serving as a valuable test-bed for responsible algorithmic misinformation detection. Published at ACM FAccT '23.
- Evaluating LLM's one-shot ability to replicate unique perspectives relevant for fact-checking, finding that they tend to exaggerate differences in opinion between groups.

Crime and Education Labs, University of Chicago **Chicago, IL**
Senior Data Scientist *Jan 2017 – June 2021*

- Developed a predictive model and data pipeline to identify individuals at high risk of experiencing domestic violence, enabling targeted support from specialized social services.
- Utilized education data to construct a predictive model that accurately identifies fourth graders at risk of dropping out before high school graduation to target with additional services.
- Developed novel causal inference technique (`ElasticSynth`) to determine effect of change in policing management on crime levels in Chicago.

Technical Skills

Programming Languages: Python, R

Database Languages: SQL

Machine Learning: PyTorch, xgboost, lightgbm

Natural Language Processing: LLM Evaluation, Amazon Bedrock, Fine-tuning, Few-shot Learning

Network Science: networkx, Gephi

Causal Inference: Matching, Synthetic Controls, Bootstrapping & Permutation Tests

Relevant Publications

- Neumann, Terrence, Maria De-Arteaga, and Sina Fazelpour. "Justice in misinformation detection systems: An analysis of algorithms, stakeholders, and potential harms." Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT). 2022.
- Neumann, Terrence, and Nicholas Wolczynski. "Does AI-Assisted Fact-Checking Disproportionately Benefit Majority Groups Online?." Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency (FAccT). 2023.
- Tanriverdi, Hüseyin, John-Patrick Akinyemi, and Neumann Terrence. "Mitigating Bias in Organizational Development and Use of Artificial Intelligence." Proceedings of the 2023 International Conference on Information Systems (ICIS). (2023).

Working Papers

- Neumann, Terrence, Sooyong Lee, Maria De-Arteaga, Sina Fazelpour, and Matthew Lease. "Diverse, but Divisive. LLMs Can Exaggerate Gender Differences in Opinion Related to Harms of Misinformation." *arxiv*. 2024.
- Neumann, Terrence, Maria De-Arteaga, Sina Fazelpour, Maytal Saar-Tsechansky, and Matthew Lease. "The Justice Implications of Ground Truth Used in AI-Assisted Fact-Checking." 2024.
- Neumann, Terrence and Bryan Jones. "PRISM: A Design Framework for Open-Source Foundation Model Safety." *arxiv*. 2024.
- Kapustin, Max, Terrence Neumann, and Jens Ludwig. Policing and management. No. w29851. National Bureau of Economic Research, 2022.

Fellowship/Awards

Brumley Fellowship in Cybsersecurity. 2023-2024: Investigating the risks associated with open-sourcing advanced AI technologies, like LLMs.

NYU Policing Project Fellow. 2023-2024: Providing data assistance and thought leadership on "Reimagining Public Safety" project, which aims to reduce violent interactions between police and civilians via alternative response.

Chicago Police Department Superintendent's Award of Merit. 2020: Awarded for work assisting Chicago Police Department develop advanced analytics capabilities.

Invited Conference Presentations

Conference on Information Systems and Technology (CIST): 2024

INFORMS Annual Meeting: 2024

INFORMS Data Science Workshop: 2024

ACM Conference of Fairness Accountability and Transparency (FAccT): 2022, 2023

Service

AAAI/ACM Conference on AI, Ethics, and Society: 2024 - Program Committee

Conference on Information Systems and Technologies (CIST): 2024 - Reviewer

INFORMS Data Science Workshop: 2024 - Reviewer

Workshop on Information Technologies and Systems (WITS): 2021 - Conference Volunteer